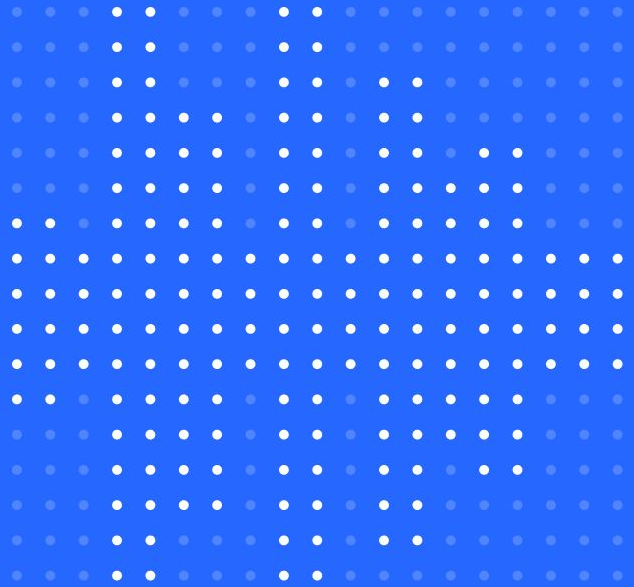
Meet Concordium

Dragos Danciulescu - DevRel Engineer
Doris Benda - Senior Software Engineer



CONCORDIUM

Useful Links



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[Follow Along, access links, presentation link]
[Official Concordium Documentation]
[Official Concordium Website]
[Concordium Github]
[Join our Discord!]
[Whitepaper]

Facts & Statistics

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- Up to 2000 TPS (scalable beyond this threshold)
- 2-4 seconds Transaction Finality
- Low transaction fee (€0.01), pegged to the Euro;
- 172,999,681 total transactions currently
- 103,173 accounts currently
- 9956 smart contracts deployed
- 11 stablecoins as Protocol-Level Tokens
- Concordium BFT (Byzantine Fault Tolerance) consensus protocol
- Open-source project

Principles of Privacy

Concordium's identity system has been designed to be privacy first and incorporates several key privacy principles.

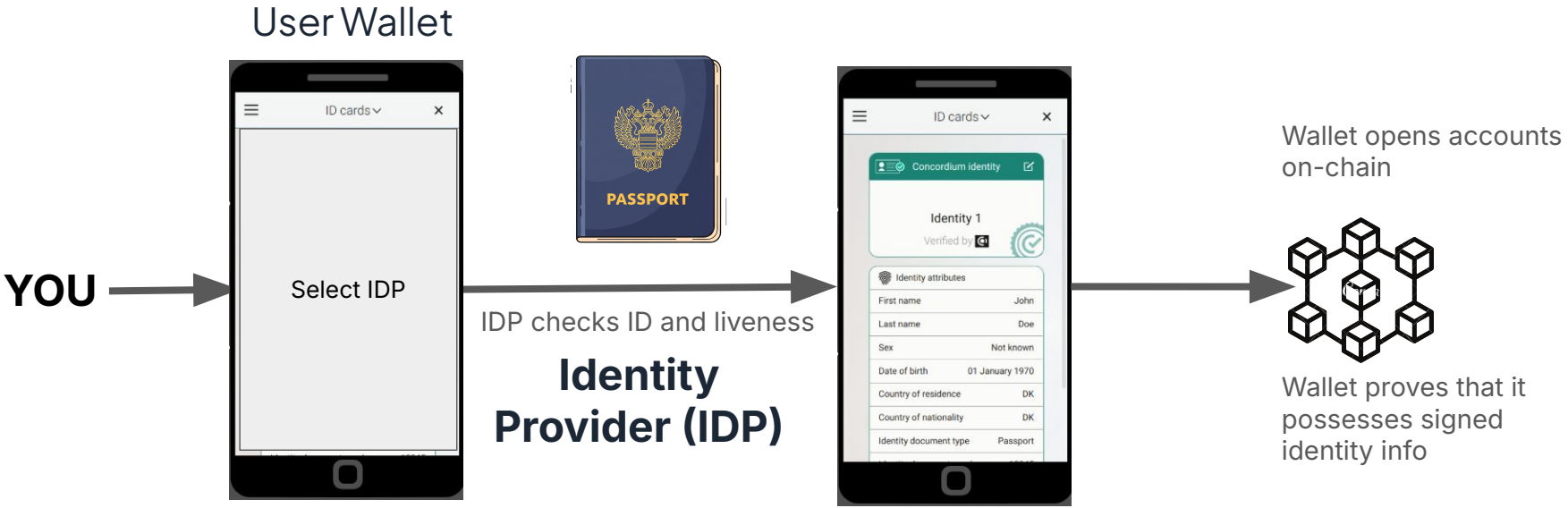
Protection of PII

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Users' **personal identifiable information (PII)** such as names, birth dates etc, is **never** available on chain, not encrypted and not unencrypted. PII is only stored locally within the user's wallet and with the Identity Provider (IDP) for compliance purposes—users retain full control and ownership of their data at all times. Concordium does not have access to, own, or store any of this user information.

Account Opening Flow

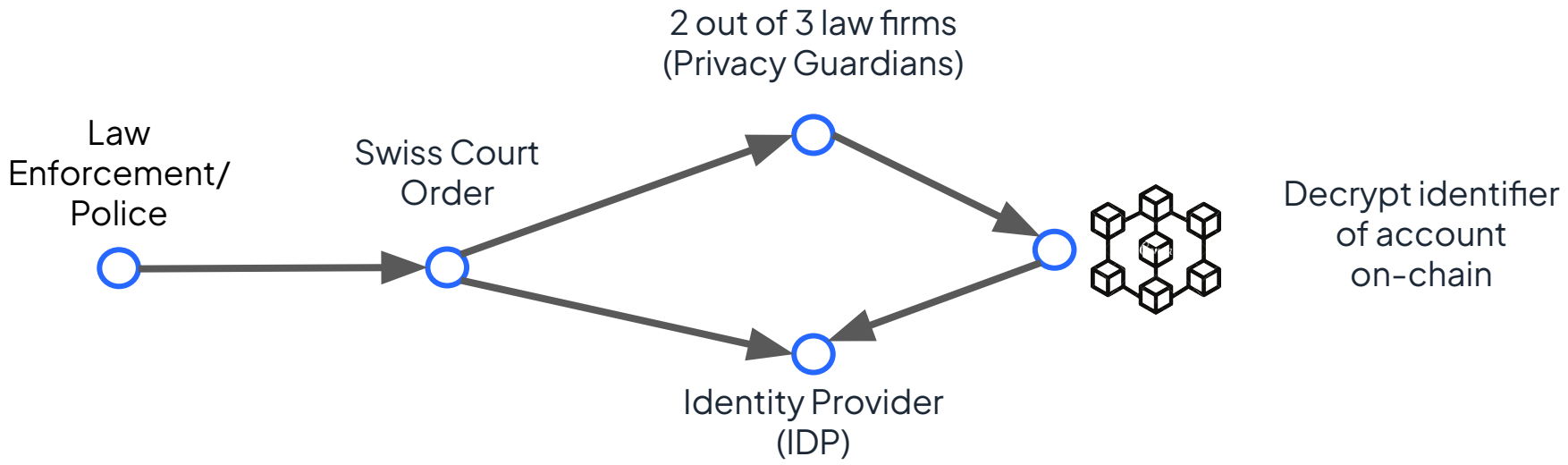
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Signed identity info stored in IDP database and wallet

Identity Disclosure Process

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Link identifier to real-world identity info to the accounts on-chain

Segregation of Power

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No single party can link a user's Identity to the accounts they have on Concordium. This can only be done through the Identity disclosure process.

- IDPs (Identity Providers) cannot identify a user's account (or on-chain wallet address). They cannot connect an identity to on-chain activity or an address.
- A single Privacy Guardian cannot decrypt a user's Identity Disclosure Data. This means they cannot access the mapping between a user's identity and their on-chain presence. In addition they do not have access to the PII which is stored within the IDP's systems.
- It is only possible to connect a user's PII identity to an account by following the identity disclosure process.

Empowered Users via Zero-knowledge (ZK) Proofs

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With verified identities in the wallets, users can prove:

- Their name is xyz (**selectively disclose** some info of their identity).
- They are older/younger than xyz (**range proofs**):
  e.g., proving they are over 18 years without revealing their birth date.
- They belong or do not belong to a group (**set/not-set membership proofs**):
  e.g., proving they are a citizen of the EU

Proof Explorer



ZK Proof Explorer Deep Dive

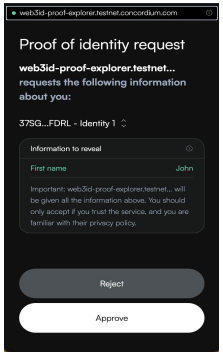
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Frontend

1. Send Request

2. Return ZK Proof



Wallet

3. Send ZK Proof

5. Proof Valid / not Valid

Blockchain

4. Look Up Cryptographic Commitment



Backend Verifier

So how hard are ZKP's for Devs?

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- No need to build a ZK dApp (plain PLTs, smart contracts, dApps exist)
- If you want to use ZK, no ZK advanced math knowledge needed by devs because:
  - Wallets do all the ZK proof generation:
    - e.g., call the wallet API endpoint `generate_proof`.
  - ZK verifier backend provided by Concordium for rapid prototyping and testing:
    - e.g., call the backend API endpoint `verify_proof`.

Protocol Level Tokens



What are PLT's?

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- Protocol-Level Tokens (PLTs) are tokens issued and managed directly on the Concordium blockchain without a need for smart contracts. PLTs provide enhanced security by reducing attack vectors, improved efficiency in execution, and direct governance controls for token issuers.
- Concordium has onboarded 4 stablecoin issuers with 11 PLTs in just the last 5 months.
- Concordium's Protocol-Level Tokens (PLTs) deliver secure, compliant, and efficient programmable assets built for today's PayFi economy — smart money, not just smart contracts.

PLT Operations

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Follow this link for code examples that showcase the following operations in Typescript and Rust:

- Mint tokens: Mint new tokens (issuer only)
- Burn tokens: Burn existing tokens (issuer only)
- Add account to allow list: Add an account to the allow list for a PLT (issuer only)
- Remove account from allow list: Remove an account from the allow list for a PLT (issuer only)
- Add account to deny list: Add an account to the deny list for a PLT (issuer only)
- Remove account from deny list: Remove an account from the deny list for a PLT (issuer only)
- Pause: Suspends balance transfer operations for a PLT (issuer only)
- Unpause: Resumes balance transfer operations for a PLT (issuer only)

Installation Prerequisites

Set up your development environment by following this guide

Developer Tools

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- Concordium Wallet for Web (Browser Wallet)
- Concordium Client (Command Line Tool)
- CCDScan (Blockexplorer)
- Public testnet node gRPC address: grpc.testnet.concordium.com
- SDKs and APIs
- Proof Explorer

Local chain setup

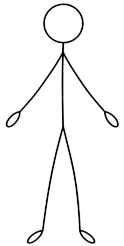
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- Full Protocol 9 local chain in minutes
- Block explorer explorer included
- Pre-funded test accounts
- Custom PLT token creation
- Github repo with guide



A very realistic problem

You



The problem

Many platforms require you to upload your ID or passport to prove nationality, exposing all your personal data.

A better way

What if you could prove you're an EU national without revealing your identity, address, or any sensitive information?

Enter: Allowlist Faucet

This demo app shows how Concordium's Protocol-Level Tokens (PLTs) protect your privacy by design.

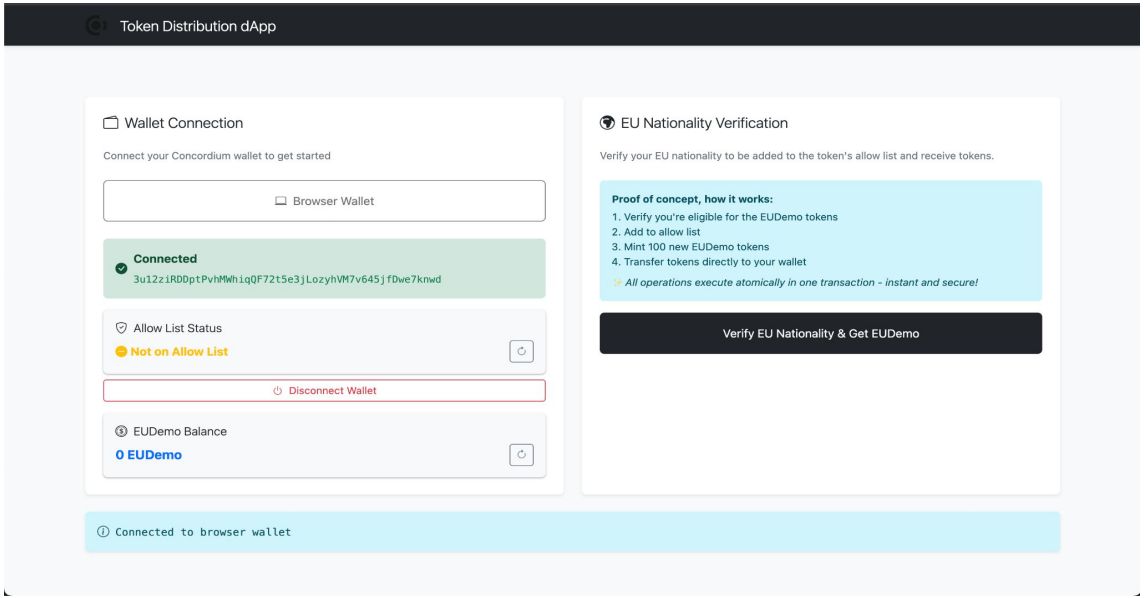
How it works

Instead of showing your ID, you generate a zero-knowledge proof that confirms only one thing: you're an EU national. Then, you’re eligible to receive tokens.

Opening the dApp

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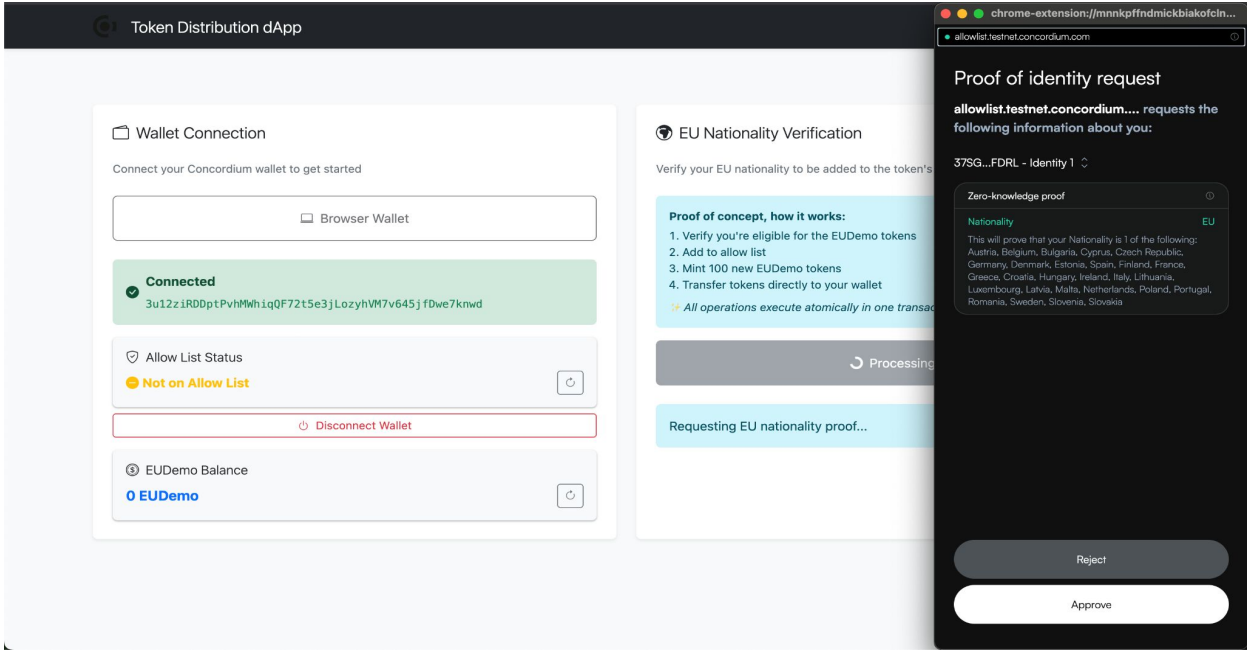
You connect your Concordium wallet to the Allowlist Faucet on testnet.



Generate the proof

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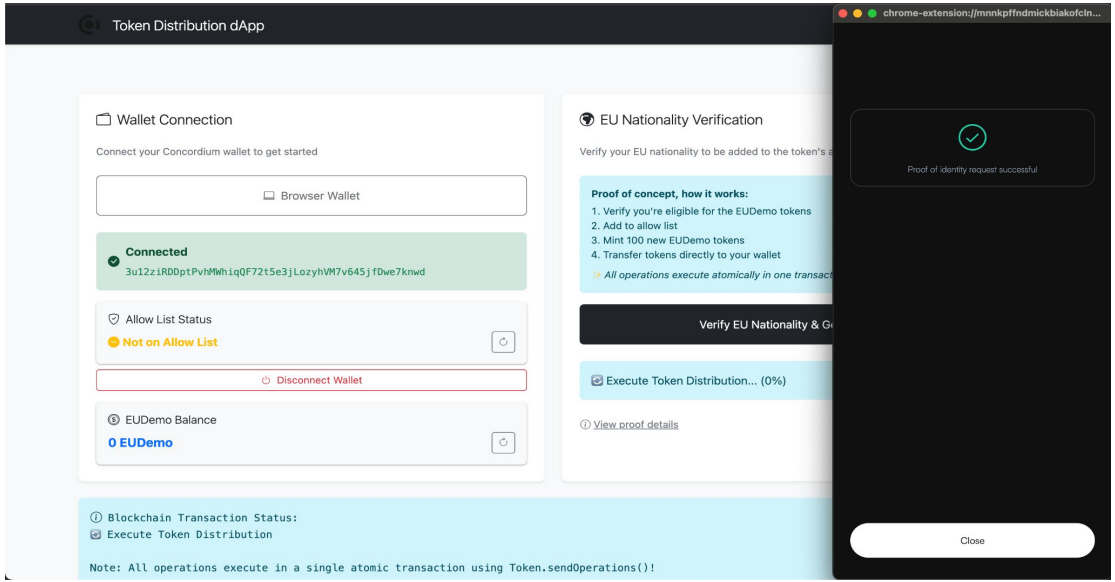
With one click, your wallet creates a ZK proof from your identity—no documents, no data sharing.



Submit and verify

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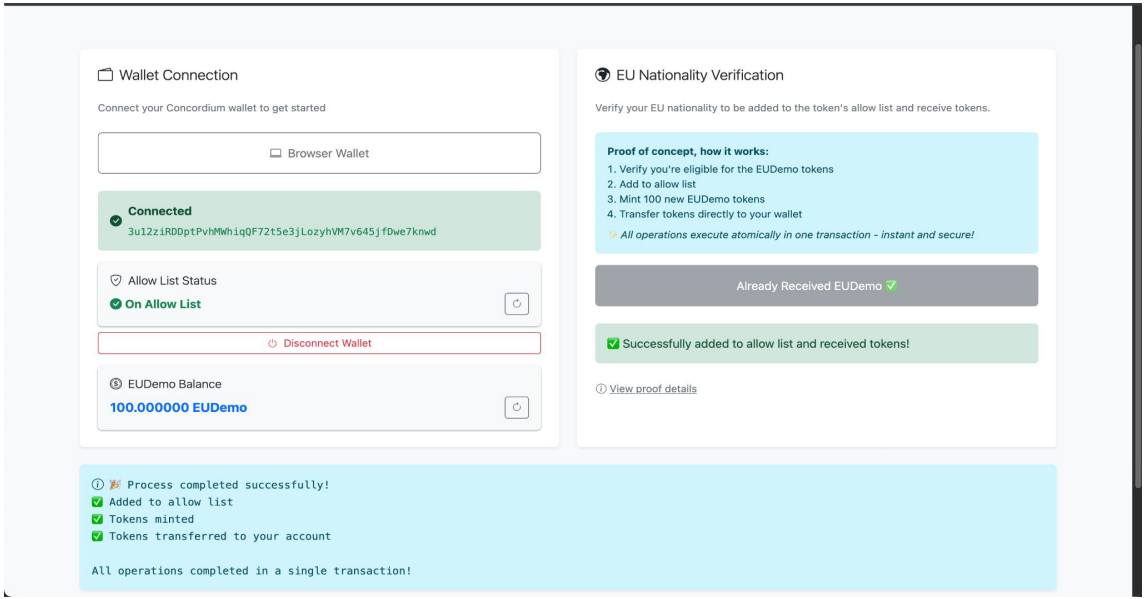
The dApp verifies your proof off-chain, confirming you're eligible without ever seeing who you are.



Your privacy intact

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You now have 100 PLTs to spend, and your personal information never left your wallet.



Under the hood articles

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Prize pools

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🥇 1st Place: $9,000 in CCD
Build on Concordium

🥈 2nd Place: $4,000 in CCD
Build on Concordium

🥉 Bonus Prize: $1,000 in CCD
for a team that didn't take part in our challenge track

Join our discord for a chance to win 500CCD
Star our github repos for a chance to win 500CCD

Find us

CONCORDIUM

Discord

Github

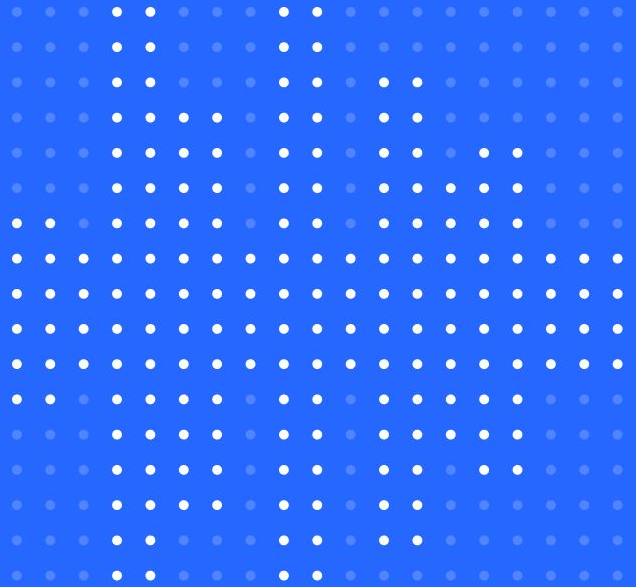
https://github.com/DOBEN
https://github.com/dragosgd

LinkedIn

www.linkedin.com/in/DorisBenda/
https://www.linkedin.com/in/dragos-gabriel-danciulescu-087552213/

We will be here for help and questions during the hackathon.

Thank you



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